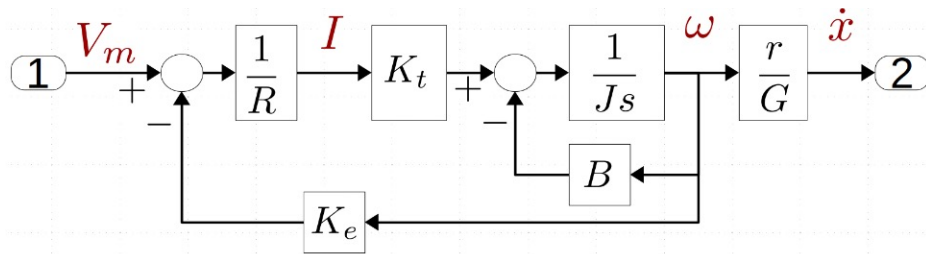


Exam - Spring 2020 (S20)

Solution

S20 Question 1

A brushless motor drives a skateboard with a person of 58 Kg. The motor is controlled with a voltage V_m between 0 and 22 volts, and the system is modelled with the following block diagram:



The following values have been found:

$K_t = 0.035 \text{ N m A}^{-1}$ (constant torque motor),

$K_e = 0.035 \text{ Vs}$ (motor back-emf constant),

$R = 0.05 \text{ } \Omega$ (motor internal resistance),

$J = 0.015 \text{ Kg m}^2$ (inertia from, among other things, person weight),

$r = 0.05 \text{ m}$ (wheel radius),

$G = 3.2$ (gearing),

I is the motor current [A], V_m is the motor voltage [V], ω is the motor angular velocity [rad s⁻¹] and $\dot{x}$ is speed [m s⁻¹].

Thus, only the friction B is missing, which is determined on the basis of a measurement at the operating point. Operating point is $\dot{x} = 3 \text{ m s}^{-1}$. At this speed the motor draws $I = 8.5 \text{ A}$ at a motor voltage of $V_m = 7.2 \text{ V}$.

At steady state, the input to integrator:

$$I_0 K_t = B \omega_0$$

and

$$\omega_0 = \dot{x}_0 \frac{G}{r} \Rightarrow$$

$$I_0 K_t = B \dot{x}_0 \frac{G}{r} \Rightarrow$$

$$B = \frac{I_0 K_t}{\dot{x}_0 \frac{G}{r}} \Rightarrow B = \frac{8.5 \times 0.035}{3 \frac{3.2}{0.05}} \Rightarrow B = 0.0016$$

Select an answer option:

☐ $B = 0.1$

☐ $B = 57.7$

☒ $B = 0.0016$

☐ $B = 0.014$

What value does B have at this operating point?

S20 Question 2

A steam engine is fed with steam through a valve.

Where:

ω is the engine angular velocity in rad/sec,

a is the valve opening - between 0 and 1,

$J = 0.23 \text{ Kg m}^2$ (inertia of steam engine and load),

$B = 0.12 \text{ N m s}$ (load and friction),

$K = 0.056 \text{ N m} / \sqrt{Pa}$ (system constant),

$P = 3 \times 10^5 \text{ Pa}$ (steam overpressure),

$H = 1600 \text{ Pa s}$ (pressure drop due to flow)

The following (simplified) relationship between valve position (a) and the steam engine angular velocity (ω) is stated:

$$J\dot{\omega} + B\omega = a K \sqrt{P - H\omega}$$

What will be the transfer function at an operating point where the motor is running at 600 rpm (revolutions per minute)?

☐ $G(s) = \frac{\omega}{a} = \frac{0.26}{s + 0.52}$

☐ $G(s) = \frac{\omega}{a} = \frac{0.11}{s + 0.52}$

☐ $G(s) = \frac{\omega}{a} = \frac{0.08}{s + 195}$

☒ $G(s) = \frac{\omega}{a} = \frac{109}{s + 0.65}$

☐ $G(s) = \frac{\omega}{a} = \frac{25}{s + 0.15}$

Operating point 600 RPM

$$\omega_0 = \frac{600 \times 2\pi}{60} = 62.8 \text{ rad/s}$$

And corresponding valve position (steady state)

$$B\omega_0 = K a_0 \sqrt{P - H\omega_0}$$

$$a_0 = \frac{B\omega_0}{K\sqrt{P - H\omega_0}} \Rightarrow a_0 = \frac{0.12 \times 62.8}{0.056\sqrt{300000 - 1600 \times 62.8}} \Rightarrow a_0 = 0.3$$

Linearizing for variables

$$\dot{\omega} \quad \omega \quad a$$

$$J\Delta\dot{\omega} + B\Delta\omega = K\sqrt{P - H\omega_0}\Delta a - 0.5\frac{K a_0 H}{\sqrt{P - H\omega_0}}\Delta\omega$$

Insert values and Laplace transform

$$0.23s\omega + 0.12\omega = 25a - 0.03\omega$$

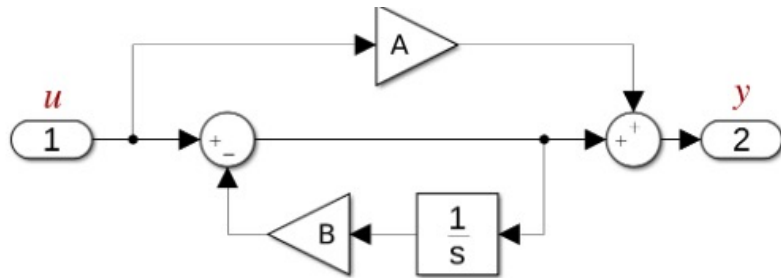
Isolate and normalize s in denominator

$$G(s) = \frac{\omega}{a} = \frac{25}{0.23s + 0.12 + 0.03}$$

$$G(s) = \frac{\omega}{a} = \frac{108.7}{s + 0.652}$$

S20 Question 3

What is the transfer function for this block diagram?



There is a sum of two branches.

The upper branch

$$G_U(s) = A$$

The lower branch is a loop, that can be reduced.

$$G_L(s) = \frac{1}{1 + \frac{B}{s}}$$

$$G_L(s) = \frac{s}{s + B}$$

In total

$$G(s) = G_U(s) + G_L(s)$$

$$G(s) = A + \frac{s}{s + B}$$

$$G(s) = \frac{A(s + B) + s}{s + B}$$

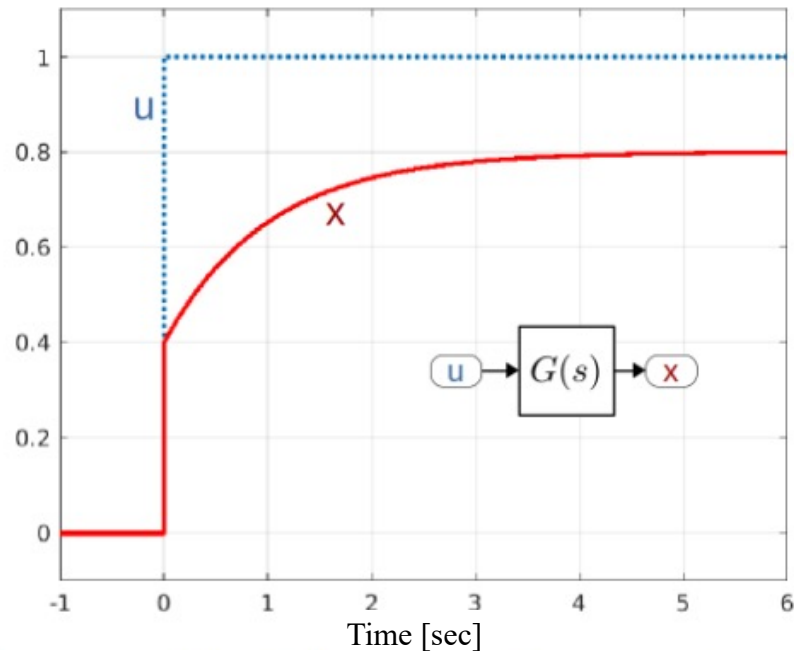
$$G(s) = \frac{(1 + A)s + AB}{s + B}$$

Select an answer option:

- ☐ $G(s) = \frac{1 + A}{Bs + 1}$
- ☒ $G(s) = \frac{(1 + A)s + AB}{s + B}$
- ☐ $G(s) = \frac{(1 + A)s}{s + B}$
- ☐ $G(s) = \frac{(1 + A)s}{s + A + AB}$
- ☐ $G(s) = \frac{1 + A}{1 + A(1 + B)s}$

S20 Question 4

A system has the following step response



What type of transfer function can provide this response?

The initial value is 0.4 and the initial value can be found as

$$x(t=0) = \lim_{s \rightarrow \infty} (s u(s) G(s))$$

And for a unit step $u = \frac{1}{s}$

$$x(t=0) = \lim_{s \rightarrow \infty} G(s) = 0.4$$

This can only be done with a transfer function

$$G(s) = \frac{b_1 s + b_0}{s + a_0}$$

And with a final value 0.8 and pole time constant ~ 1 sec

$$G(s) = \frac{0.4s + 0.8}{s + 1}$$

Select an answer option:

☐ $G(s) = \frac{b_0}{s + a_0}$

☐ $G(s) = \frac{b_0}{s^2 + a_1 s + b_0}$

☒ $G(s) = \frac{b_1 s + b_0}{s + a_0}$

☐ $G(s) = \frac{s + b_0}{s^2 + a_1 s + a_0}$

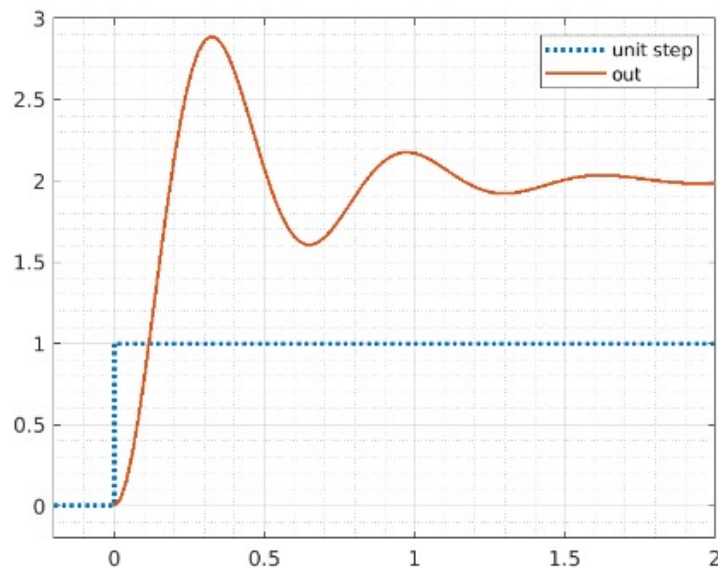
S20 Question 5

A system with the transfer function

$$G(s) = \frac{200}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

and $\omega_n = 10 \text{ rad s}^{-1}$

has the following unit step response:



Where the peak value is 2.9.

What value has ζ in the transfer function?

Damping factor ζ determines the overshoot

The overshoot is:

$$M_P = \frac{0.9}{2} = 0.45 \text{ (45\%)}$$

This corresponds to ζ 0.2 (see slides).

Or it can be calculated from

$$M_P = e^{\frac{-\pi\zeta}{\sqrt{1-\zeta^2}}}$$

By using MATLAB:

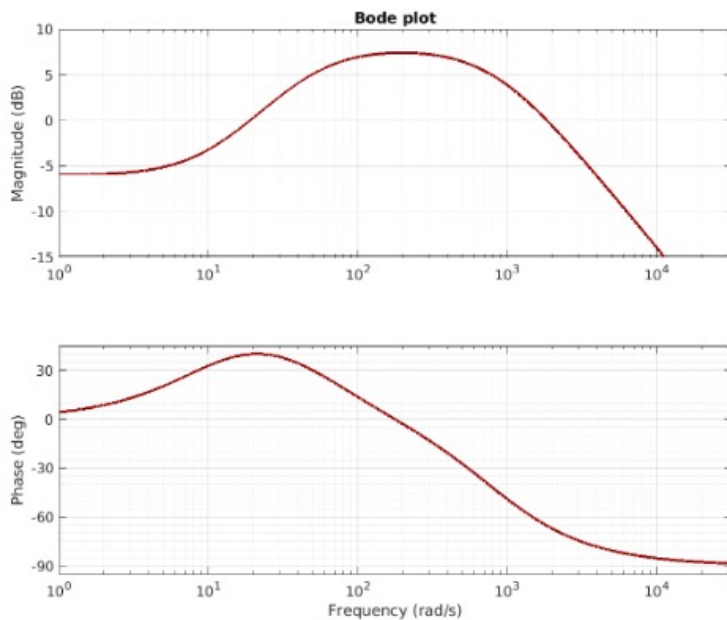
$$\zeta = 0.2$$

Select an answer option:

- ☐ $\zeta = 0.1$
- ☒ $\zeta = 0.2$
- ☐ $\zeta = 0.4$
- ☐ $\zeta = 0.7$

S20 Question 6

A transfer function has the following bode plot



Where does the transfer function have poles and zeros?

Just poles and zeros in left halfplane – from shown possibilities i.e., amplitude and phase turn up with a zero and down with a pole.

First a zero (lowest frequency) about 10 rad/s ($s = -10$).
Then first pole about 50 rad/s, as the influence on the phase is equal at about 20 rad/s, so first pole $s = -50$

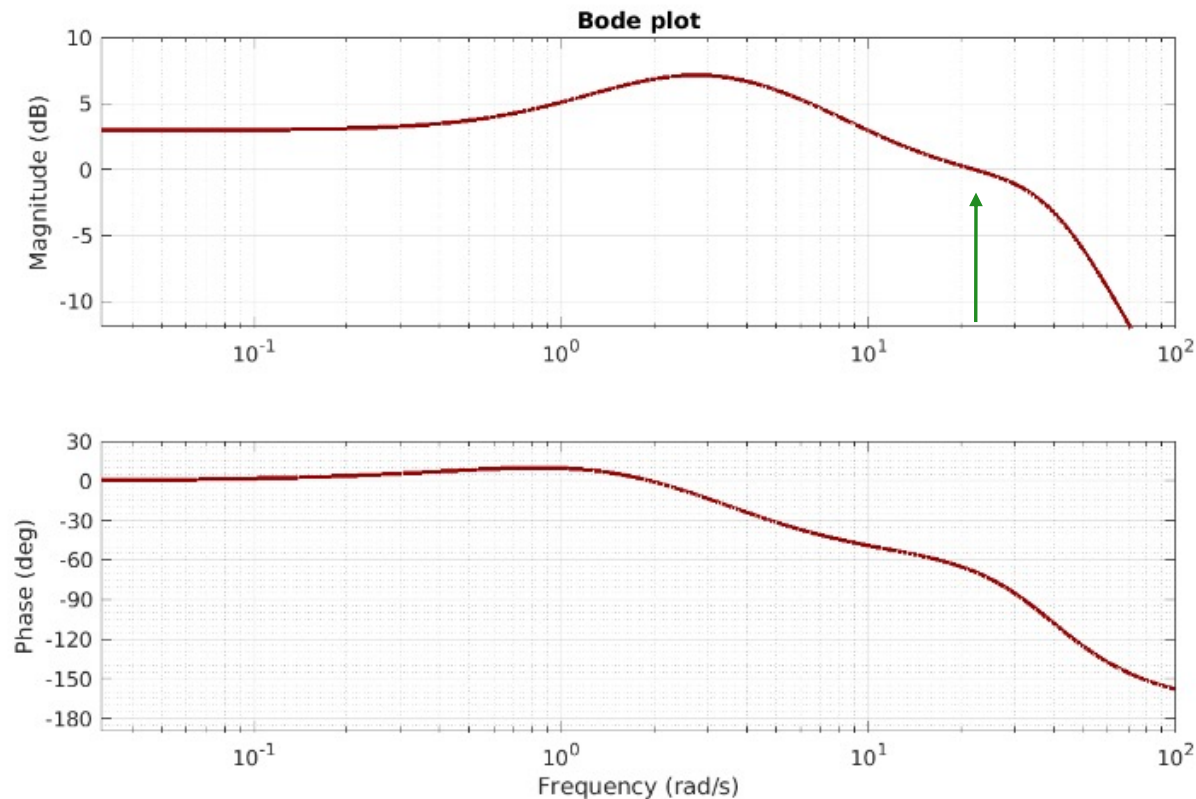
Then one more pole only as end phase is -90 deg, at around 1000 rad/s (phase around -45 degrees)

Select an answer option:

- ☒ Poles $s = -50$ and $s = -1000$
- ☐ Poles $s = -10$ and $s = -1000$
- ☐ Poles $s = -50$ and $s = -200$
- ☒ Zeros $s = -10$
- ☐ Zeros $s = -10$ $s = -200$
- ☐ Zeros $s = -10$, $s = -1000$ $s = -5000$
- ☐ No zero points

S20 Question 7

What is the bandwidth (BW) for this (closed loop) transfer function?



The bandwidth is where the amplitude is fallen to -3 dB compared to the low frequency asymptote.

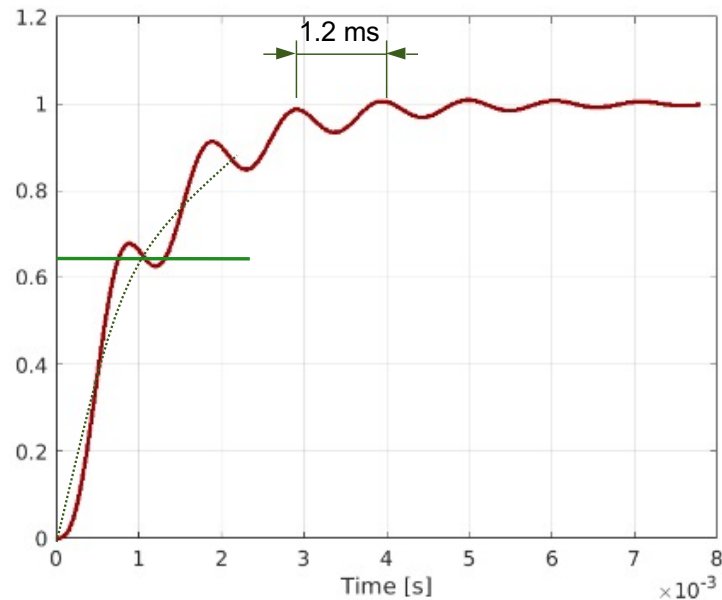
This starts at +3 dB, so crossing of 0 dB first time is the bandwidth.

Select an answer option:

- ☐ $\text{BW} = 1.4 \text{ rad s}^{-1}$
- ☒ $\text{BW} = 22 \text{ rad s}^{-1}$
- ☐ $\text{BW} = 10 \text{ rad s}^{-1}$
- ☐ $\text{BW} = 39 \text{ rad s}^{-1}$
- ☐ $\text{BW} = 2.7 \text{ rad s}^{-1}$

S20 Question 8

A closed loop system has the following unit step response



Which of the following closed loop amplitude Bode plot fits with this response?

Step analysis:

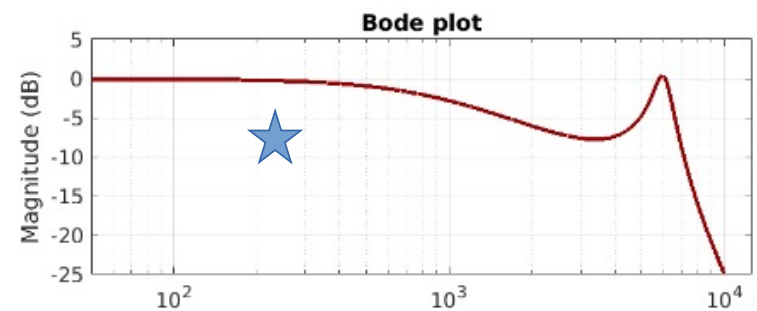
Ignoring the fast damped frequency, it looks like a 1st order pole system. The amplitude rises fast at time 0 and levels out exponentially towards 1, and it has reached 63% after about 1ms.

A 1st order break frequency of 1000 rad/s.

The damped fast frequency must correspond to an amplitude peak at this frequency.

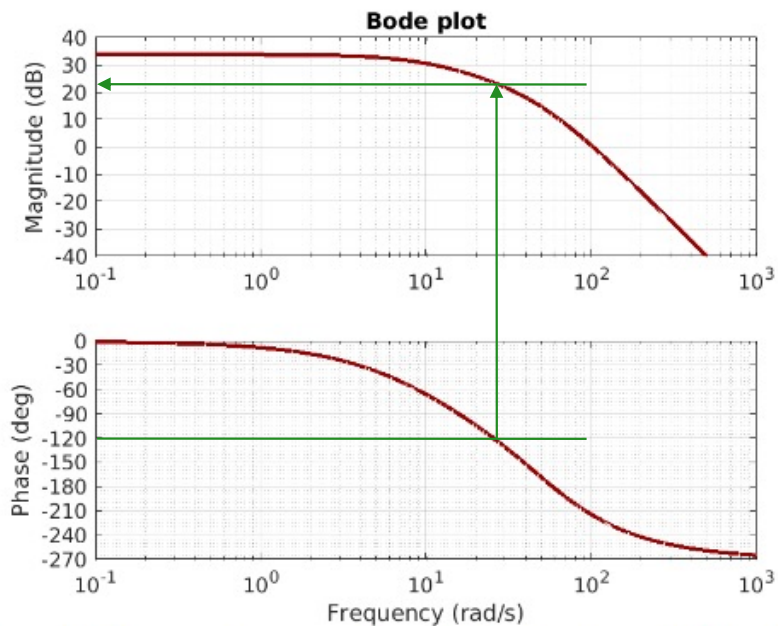
$$T \approx 1.2 \times 10^{-3} \text{ sec}$$

$$\omega_n = \frac{2\pi}{T} \Rightarrow \omega_n \approx \frac{2\pi}{1.2 \times 10^{-3}} = 5236 \text{ rad/sec}$$



S20 Question 9 - design 1

A system to be regulated has the following Bode plot



You are asked to design a P-controller to achieve a fast and stable system. What value should K_p have?

P controller design.

With phase margin of 60 degrees, the crossover frequency should be

$$\gamma_M = 60^\circ$$

$$\omega_c \approx 25 \text{ rad/sec}$$

The magnitude at this frequency is

$$|G(\omega = 25)| \approx 23 \text{ dB}$$

K_p must then be -23 dB

$$K_P = -23 \text{ dB}$$

$$K_P = 10^{\frac{-23}{20}} = 0.0708$$

Select an answer option:

☒ $K_P = 0.06$

☐ $K_P = 26$

☐ $K_P = 16$

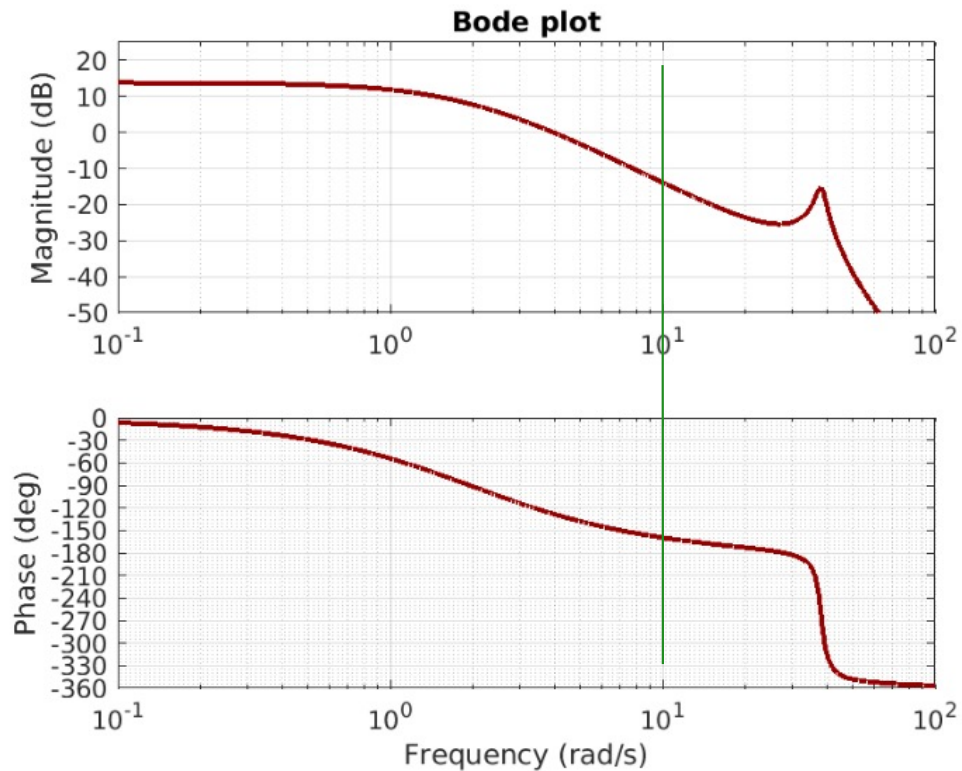
☐ The system does not become stable with a P-controller

☐ $K_P = 100$

S20 Question 10 - design 2

A system to be regulated has the following transfer function and Bode Plot

$$G(s) = \frac{28880}{s^4 + 7.04s^3 + 1460s^2 + 5788s + 5776}$$



A crossover frequency is required $\omega_c = 10$ rad / sec

To improve the phase margin, a Lead term with $\alpha = 0.1$ is designed.

What will be the zero point frequency of the Lead term?

Lead design

$$\tau_d = \frac{1}{\omega_c \sqrt{\alpha}} \Rightarrow \tau_d = \frac{1}{10\sqrt{0.1}} = 0.32 \text{ sec}$$

As is the time constant for the Lead zero

$$\omega_z = \frac{1}{\tau_d} \Rightarrow \omega_z = \frac{1}{0.32} = 3.1 \text{ rad/sec}$$

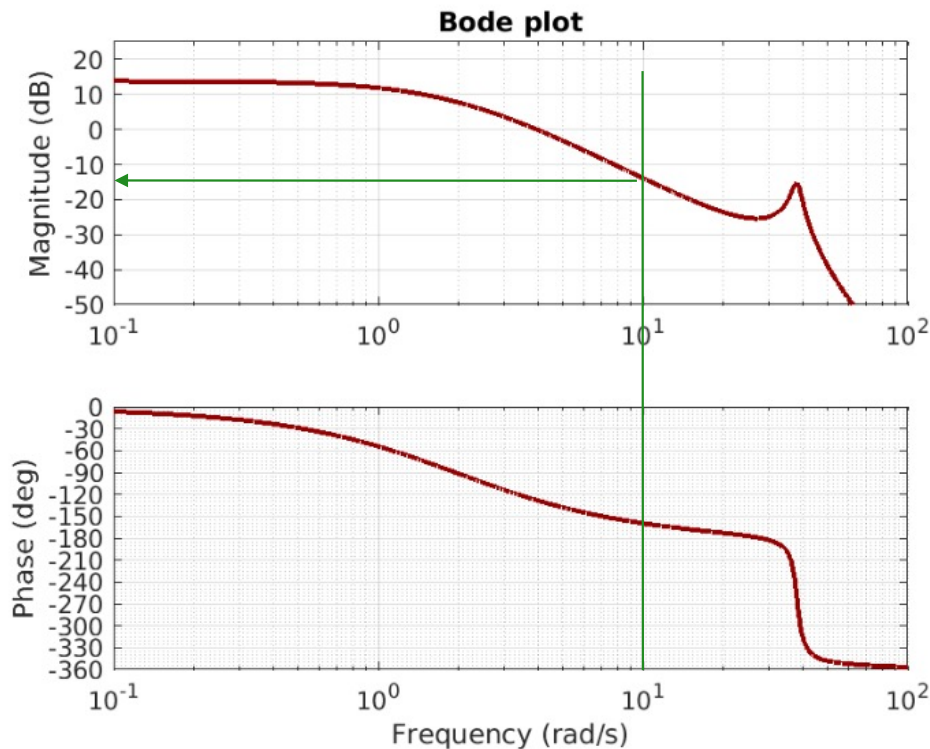
Select an answer option:

- ☐ $\omega = 0.32$ rad/ sec
- ☒ $\omega = 3.2$ rad/ sec
- ☐ $\omega = 1$ rad/ sec
- ☐ $\omega = 6.5$ rad/ sec
- ☐ $\omega = 32$ rad/ sec

S20 Question 11 - Design 2

A system to be regulated has the following transfer function and Bode plot

$$G(s) = \frac{28880}{s^4 + 7.04s^3 + 1460s^2 + 5788s + 5776}$$



A crossover frequency of $\omega_c = 10$ rad/sec is desired.

To improve the phase margin, a Lead term with $\alpha = 0.1$ is planned.

When this Lead term is part of a P-Lead regulator, what value should K_p have?

The Lead term increases the gain at the intended crossing frequency by

$$C_d(s) = \frac{\tau_d s + 1}{\alpha \tau_d s + 1}$$

The magnitude at the center frequency is

$$|C_d(\omega = 10)| = \frac{1}{\sqrt{\alpha}} \Rightarrow |C_d(\omega = 10)| = 3.2 \text{ (10 dB)}$$

Magnitude with Lead at crossing frequency is then

$$|GC_d| = -14 \text{ dB} + 10 \text{ dB} = -4 \text{ dB}$$

$$K_P = 4 \text{ dB} = 1.58$$

Select an answer option:

☒ $K_P = 1.5$

☐ $K_P = 5$

☐ $K_P = 14$

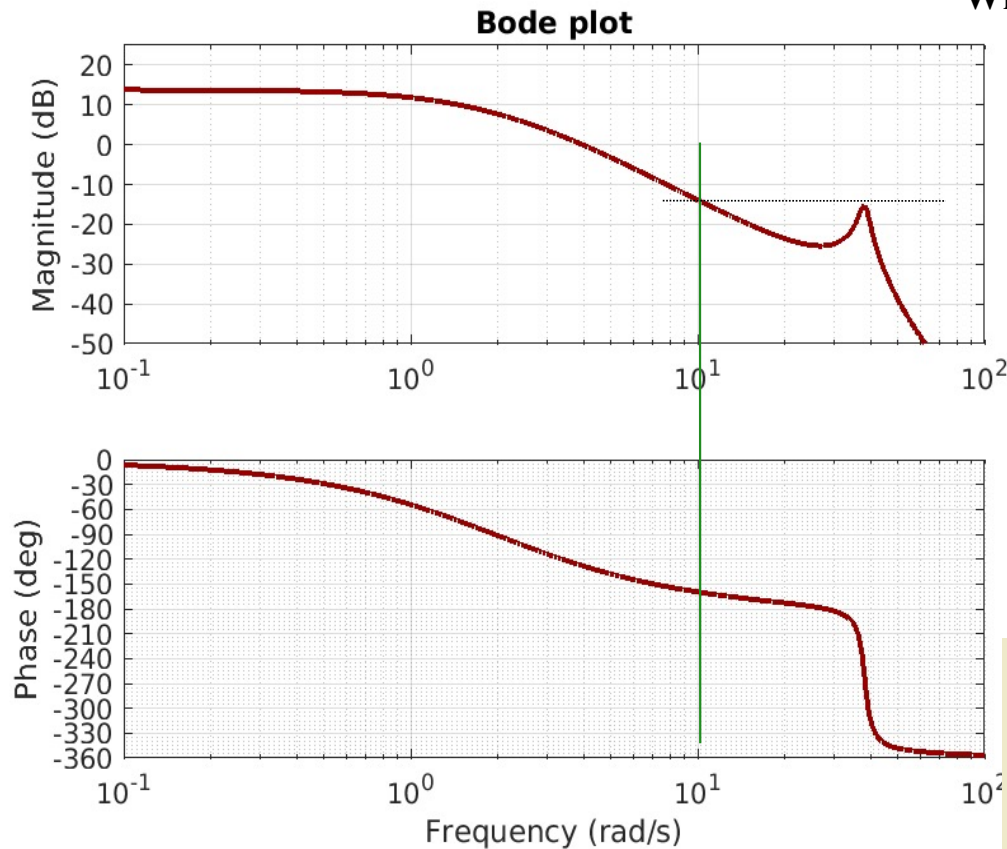
☐ $K_P = 0.76$

Could also be solved by MATLAB

```
G = tf([28880],[1 7.04 1460 5788 5776]);
wc = 10;
al = 0.1;
td = 1/(wc*sqrt(al));
Gd = tf([td 1],[td*al 1]);
[M,P] = bode(G*Gd, wc);
kp0 = 1/M
```

S20 Question 12 - Design 2

A crossover frequency of $\omega_c = 10$ rad/sec is desired
To improve the phase margin, a Lead term with $\alpha = 0.1$ is planned
Will the system be stable with this controller?



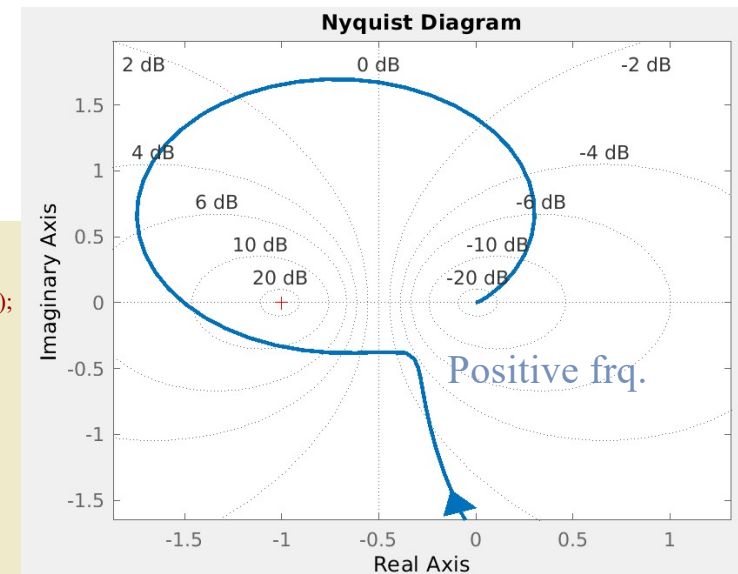
The Lead term increases the gain with frequency around the Lead center freq. and then also the magnitude peak at ~ 40 rad/sec. This peak will then be higher than 0 dB, where the phase is below -180 , so not stable.
- You can check this also in MATLAB

Select an answer option:

- ☐ Yes
☒ No

Could also be solved by MATLAB

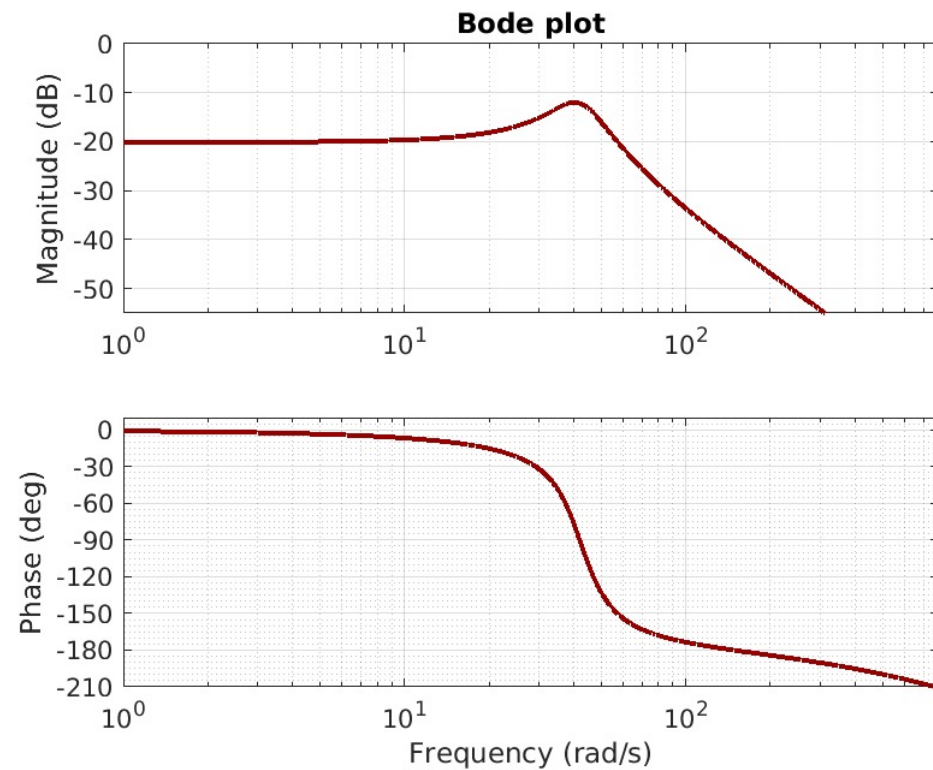
```
G = tf([28880],[1 7.09 1460 5788 5776]);
wc = 10;
al = 0.1;
td = 1/(wc*sqrt(al));
Gd = tf([td 1],[td*al 1]);
[M,P] = bode(G*Gd, wc);
kp0 = 1/M
% Nyquist plot
figure(922)
nyquist(G*Gd*kp0)
```



S20 Question 13 - Design 3

A system to be regulated has the following transfer function and Bode plot

$$\frac{229320}{s^3 + 1317s^2 + 23604s + 2.29 \times 10^6}$$



The system has 3 poles, but are any of them complex?

With poles only, complex poles are needed to get a magnitude peak.

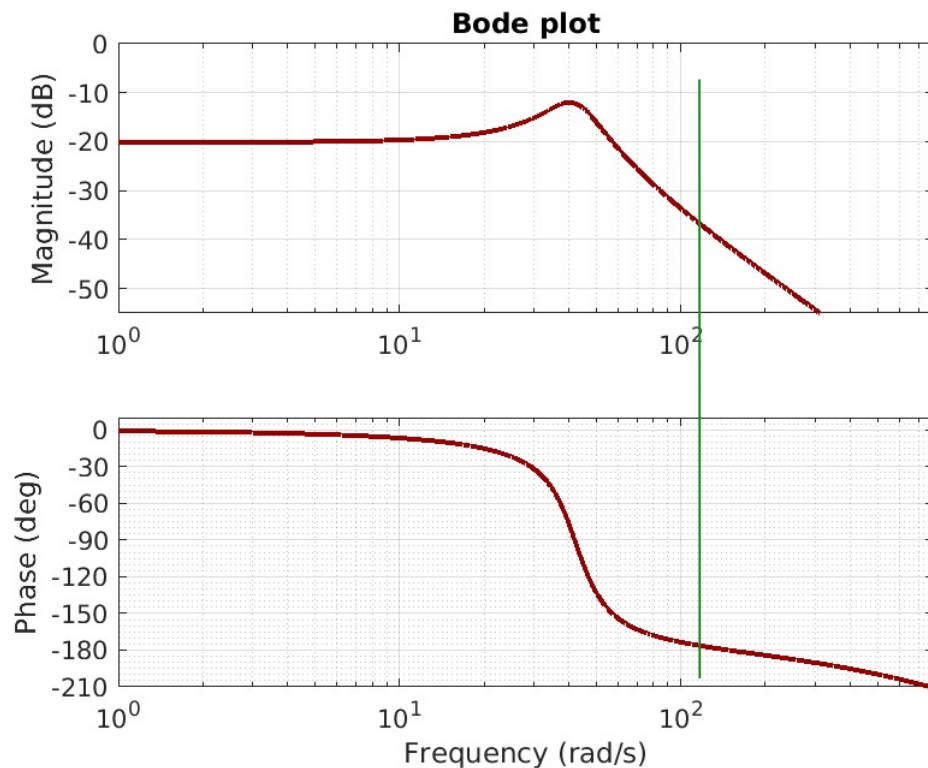
Select an answer option:

- ☒ Yes
☐ No

S20 Question 14 - Design 3

A system to be regulated has the following transfer function and Bode plot

$$\frac{229320}{s^3 + 1317s^2 + 23604s + 2.29 \times 10^6}$$



A controller must be designed for the system.

An I-link should eliminate the static error for a step input, and a Lead-term should be able to cause the crossover frequency to be at a higher frequency than the amplitude hump - especially if a small phase margin is acceptable. The following PI-Lead design parameters are selected:

$$N_i = 3, \alpha = 0.1, \gamma_M = 40^\circ$$

What value should K_p have?

Normal design job

$$\varphi_{PI} = -90 + \arctan N_i \Rightarrow \varphi_{PI} = -90 + \arctan(3) = -18.4^\circ$$

$$\varphi_m = \arcsin \frac{1 - \alpha}{1 + \alpha} \Rightarrow \arcsin \frac{1 - 0.1}{1 + 0.1} = 54.94^\circ$$

So, the crossing frequency should be where the phase is

$$\angle G(\omega_c) = -180 + \gamma_m - \varphi_m - \varphi_{PI}$$

$$\angle G(\omega_c) = -180 + 40 - 54.94 + 18.4 = -176.5^\circ$$

$$\omega_c = 122 \text{ rad/sec}$$

$$\tau_d = \frac{1}{122\sqrt{0.1}} = 0.026 \text{ sec}, \quad \tau_i = \frac{3}{122} = 0.025 \text{ sec}$$

I would use MATLAB for the rest

```
G = tf([229320],[1 1317 23604 2.29e6])
td = 0.026;
ti = 0.025;
al = 0.1;
Gd = tf([td 1],[al*td 1]);
Gi = tf([ti 1],[ti 0]);
[M,P] = bode(G*Gi*Gd,122)
Kp = 1/M
```

Select an answer option:



$K_P = 23$

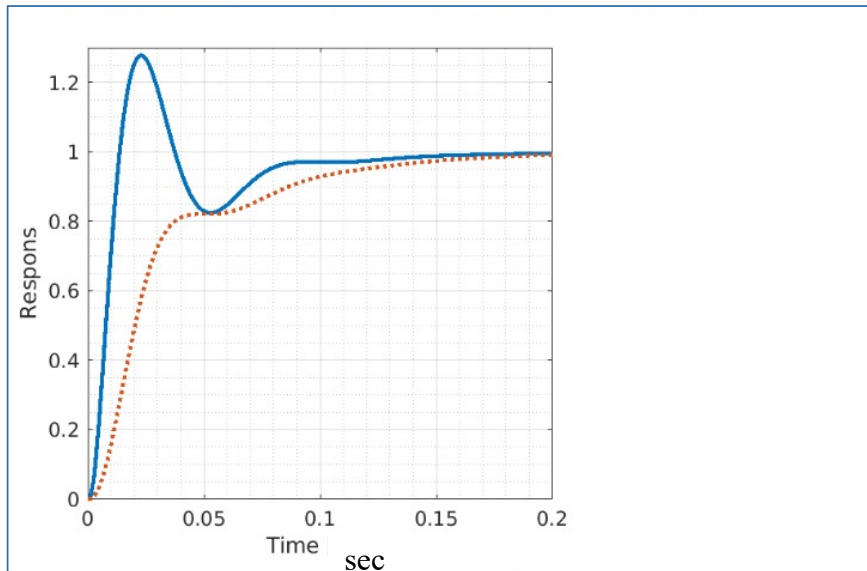
☐ $K_P = 6$

☐ $K_P = 1.1$

☐ $K_P = 190$

S20 Question 15 - Design 3

The following closed loop unit-step response is obtained with either the lead term in the forward or back branches.



Which of the couplings fits with the blue (solid) curve?

A Lead enhances change, and thus, if it is in the forward branch, it will enhance the step from ref-input. This gives a steeper rising edge than it would have with the same lead in the feedback branch.

The blue curve must therefore be with the Lead in the forward branch.

Select an answer option:

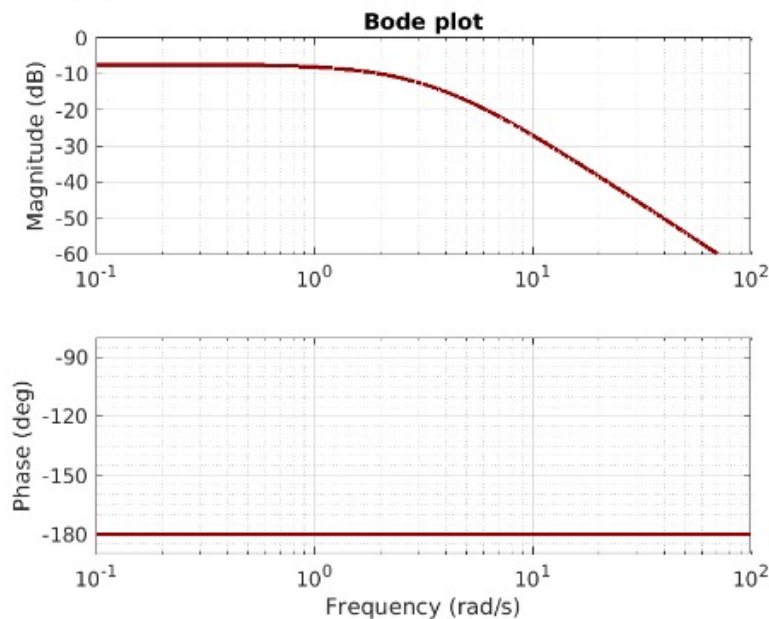
- ☒ The blue (fully drawn) curve is with the Lead term in the forward branch
- ☐ The blue (fully drawn) curve is with the Lead term in the back branch

S20 Question 16 - Stability

A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



Does the system have poles in the right hemisphere?

Sign change or missing terms in the denominator is an indication of poles in the right half-plane.

(at low frequencies ($s \rightarrow 0$) the *sign* is negative (i.e., -180 degrees)

At high frequencies, the s^2 also gives -180 degrees.)

The denominator can be factorized to

$$G(s) = \frac{5}{(s + \sqrt{11.5})(s - \sqrt{11.5})}$$

One pole in the left half-plane and one in the right.

Select an answer option:

☐ No

☒ 1 pole in the right half plane

☐ 2 poles in the right half plane

Or MATLAB

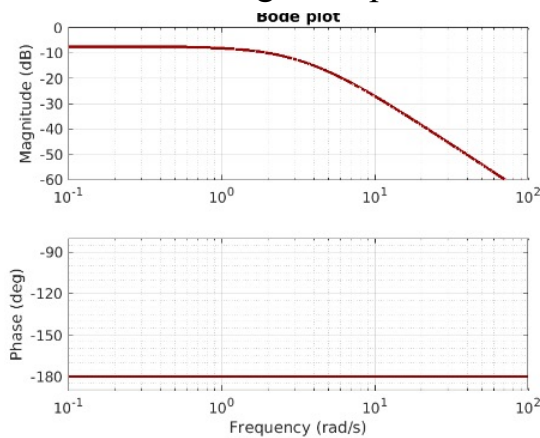
```
G = tf([5],[1 0 -11.5])
pole(G)
ans =
    3.3912
   -3.3912
```

S20 Question 17 - Stability

A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



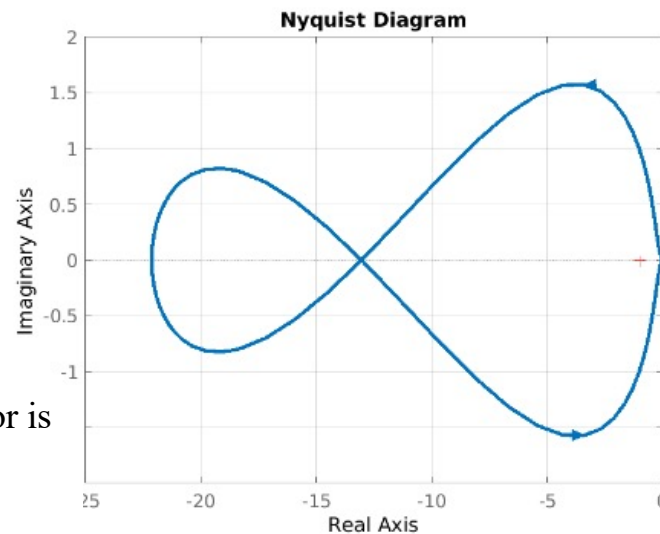
Based on the Bode plot, a P-lag-lead regulator is designed with the following parameters:

$$\omega_c = 20 \text{ rad/sec}, \alpha = 0.1, N_i = 4, \beta = 2$$

It provides the following controller components

$$C_d = \frac{0.16s + 1}{0.016s + 1} \quad C_i = \frac{0.2s + 1}{0.4s + 1} \quad K_p = 51$$

Overall, the open loop provides the following Nyquist plot



With 1 pole in the right half-plane, the Nyquist curve must encircle the -1-point one time counterclockwise (CCV) to produce a stable closed loop. (from $\omega = -\infty$ to $\omega = +\infty$)

This is the case here – following the arrows.

Select an answer option:

☒ Yes

☐ No

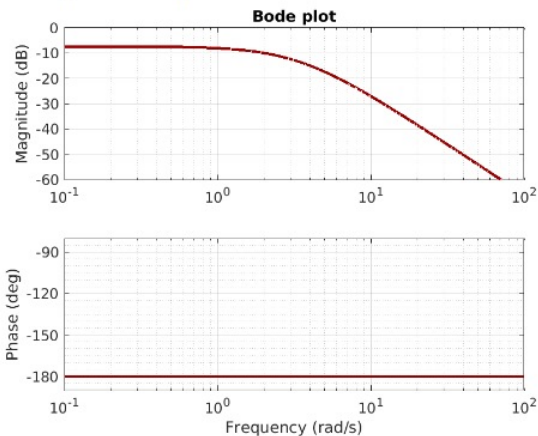
Will closed loop with this regulator be stable?

S20 Question 18 - Stability

A system is modelled with the following transfer function

$$G(s) = \frac{5}{s^2 - 11.5}$$

with the following Bode plot



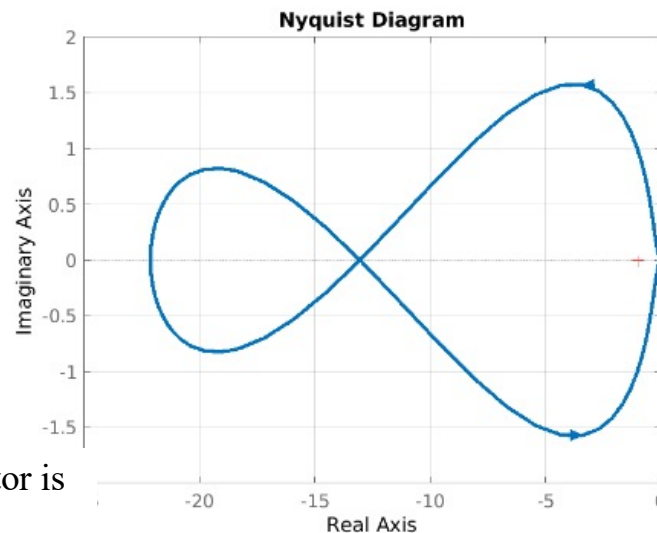
Based on the Bode plot, a P-lag-lead regulator is designed with the following parameters:

$$\omega_c = 20 \text{ rad/sec}, \alpha = 0.1, N_i = 4, \beta = 2$$

It provides the following controller components

$$C_d = \frac{0.16s + 1}{0.016s + 1} \quad C_i = \frac{0.2s + 1}{0.4s + 1} \quad K_p = 51$$

Overall, the open loop provides the following Nyquist plot



Will the system be stable if P-gain is reduced to $K_p = 1$ (and other parts - C_i and C_d - remain unchanged)?

If K_p is reduced to

$$K_p = 1$$

This corresponds to dividing the Nyquist plot with 51, and as the most negative value is -22, then this would be reduced to -0.44.

Then -1 is not encircled, and thus the closed loop will be unstable.

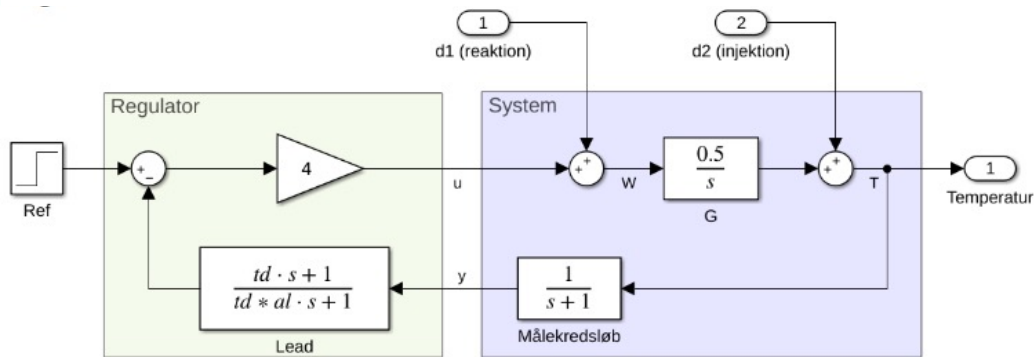
Select an answer option:

☐ Yes

☒ No

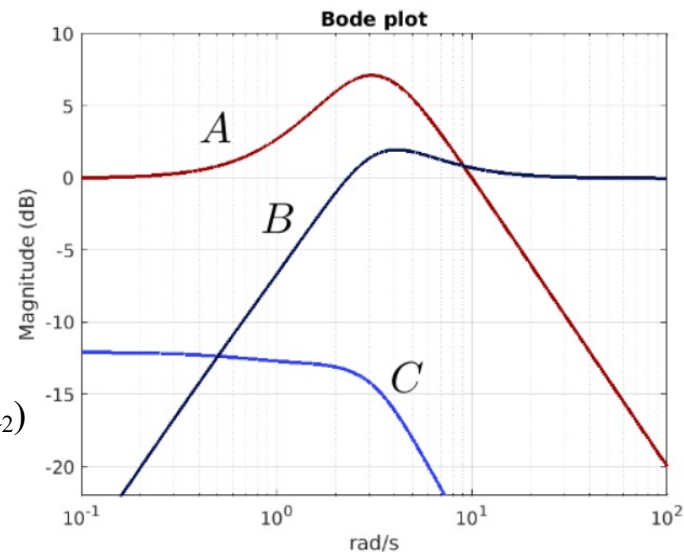
S20 Question 19 - Sensitivity

An insulated tank (bio-reactor) with in-P-Lead temperature regulator is modelled as follows:



There are two disturbances, d_1 which is due to the development of chemical energy in the tank [Watt] and d_2 which is a change in temperature due to the replacement of part of the contents of the tank [°C].

The sensitivity to temperature change due to reaction (d_1) and due to replacement of content (d_2) is shown in the following amplitude Bode plot (together with the closed loop transfer function):



Which curve corresponds to the d_1 disturbance?

The transfer function from a disturbance (end ref) describes the sensitivity for this input.

From ref, the goal is 0dB, i.e., curve A

There is an integrator (G), so a disturbance after this (d_2) will have no steady state error, i.e., gain = 0 for frequency zero.

d_2 must therefore be curve B

The last curve C must then be d_1 .

As W must be 0 in steady state (input to an integrator), then u must be $-d_1$.

And with $K_p = 4$, then the error must be 25% or -12dB.

Select an answer option:

☐ A

☐ B

☒ C